Why Hydrant Flow Tests Matter

Hydrant flow tests are conducted to determine pressure and flow-producing capabilities at any location within a distribution system. Test data may be used for a variety of applications:

- Determining how much water is available for fighting fires
- Detecting where excess head loss occurs and where closed valves are likely
- Calibrating friction factors for pipes in a hydraulic model
- Creating dynamic boundary conditions in a hydraulic model

Hydrant flow tests are based on guidelines from several industry organizations, including the American Water Works Association (AWWA), the National Fire Protection Association (NFPA), and the Verisk Insurance Services Office (ISO). This calculator is based on equations included in AWWA M32 and M17 as well as NFPA 291. The equation is defined below.

$$Q = Q_t \times ((P_s - P) \div (P_s - P_r))^{0.54}$$

Q = Projected flow at a chosen residual pressure (gpm)

Q_t = Total test flow during the test (gpm)

P_s = Static pressure with no flow (psi)

P_r = Residual pressure during the test (psi)

P = Residual pressure at which flow is projected (psi); e.g. 20 psi for available fire flow

This dashboard helps you analyze hydrant flow test data and make informed decisions

Hydrant Flow Test Calculator

PRESSURE HYDRANT

Static Pressure (psi): 60

Residual Pressure (psi): 50

FLOW HYDRANT(S)

+ Add Hydrant

Hydrant 1

Diameter (in): 2.5

Inlet Coefficient: Rounded (0.9)

Pitot Pressure (psi): 40

Flow: **1,061 gpm**

RESULTS

Flow per Hydrant (gpm)

Hydrant 1 — 1,061

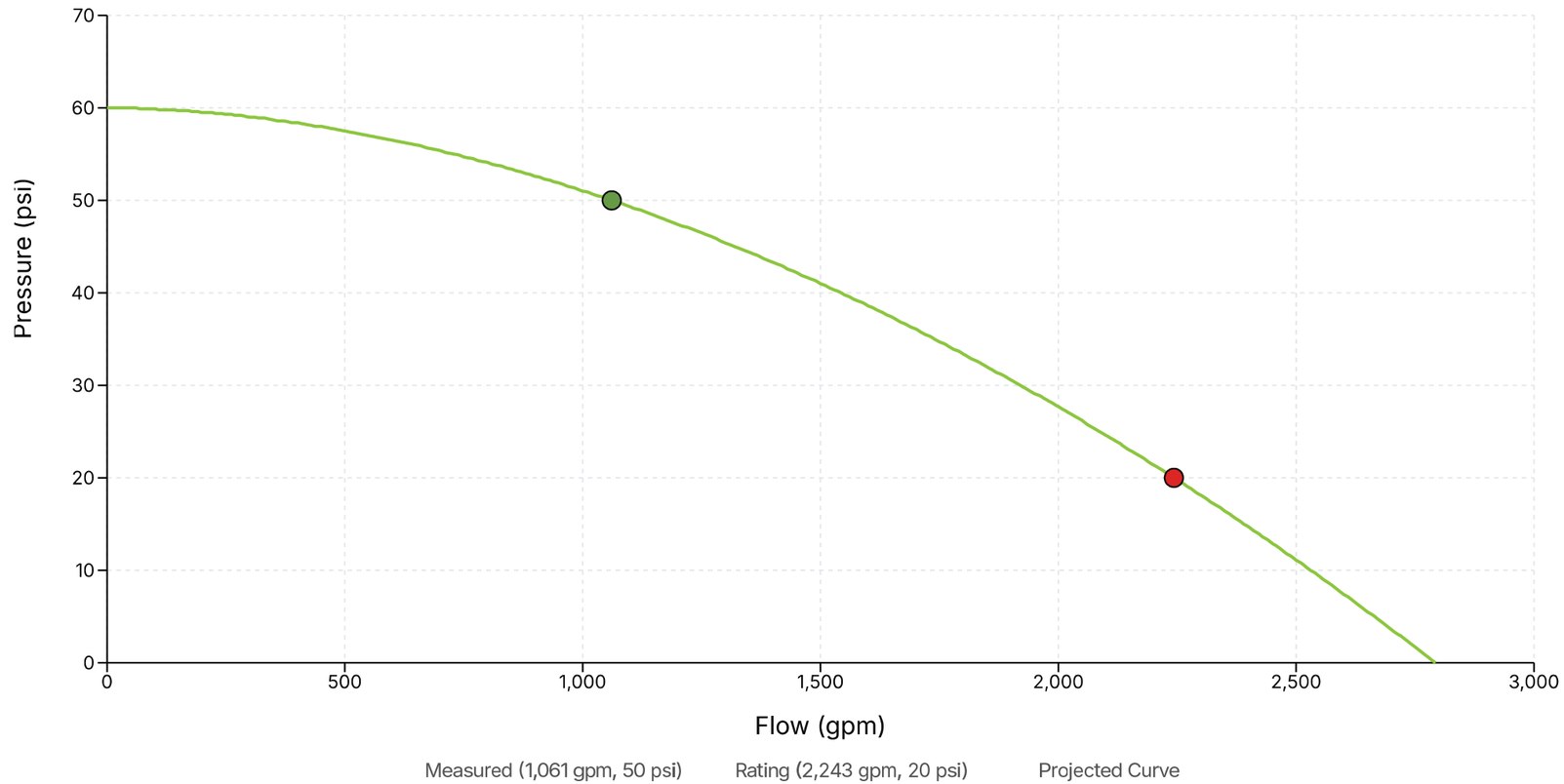
Total Test Flow

1,061 gpm

Projected Flow at Static Pressure
(at 20 psi residual, NFPA 291)

2,243 gpm

Pressure–Flow Curve


Pressure (psi)
70
60
50
40
30
20
10
0
0
500
1,000
1,500
2,000
2,500
3,000
Flow (gpm)
Measured (1,061 gpm, 50 psi)
Rating (2,243 gpm, 20 psi)
Projected Curve